Reseach Assignment of
Exercise 1 The language of Business Analytics

SECTION 1 - Foundational Concepts of Analytics

1) Business Analytics - it is the practice of using data, statistical analysis, and ~~technology~~ technology to examine business performance, dicover insights and drive data-informed decisions. More like turning raw numbers from your ~~bu~~ business into actions that make money or save it.

Example in Tembisa / kasi context
A taxi rank kota shop tracks daily sales in a notebook.
Business Analytics = putting that info into excel, identifying that "payday = 3x chips sales" then ordering ~~stock~~ extra stock a day before payday is important to not lose sales on payday.

2) Data Analytics - it is the process of examining raw data to find, answer questions and draw conclusions. It is more like taking messy data and converting it into useful information.

Example - Data Analytics = analyse weather in Soweto → finds out/discover that it rains a lot in January

Business Analytics → use the rain data → decide the car wash should run "rainy day discounts" in December to be able to log in customers before January plunge.

{ Answers above are from GARTNER & IBM / MICROSOFT
Source - Provost, f. & Fawcett
Authour - Foster Provost & Tom Year - 2013 }

3) Business Intelligence (BI)

it is a process which involves technology-driven process. to ~~process~~ tran-form raw data into meaning-ful information that Supports Strategic decision making in organisations

BI ~~solves~~ Example / Scenario

- Managers with inaccurate or incomplete info will tend on average, to make worse decisions than if they had better info.

- BI - solves the matter above by collecting, transforming and analyzing data, showing it with dashboards and visuals to aid decision making. BI tools automate much of this process allowing companies to make quicker, more accurate decisions by identifying trends, uncovering ~~incost~~ incosistencies and providing actionable insights.

4) Data-Driven Decision Making

Refers to the use of data to inform organizational decisions and approach.

Example / Scenario

HR teams apply talent analytics to work-force planning, ~~perfo~~ Performance evaluation and retention strategies.

Ops and Supply Chain - may also use customer data and machine learning to forecast demand, schedule ~~equ~~ equipment maintainance and to also manage inventory.

5) Big Data - data that is too large, fast or too complex that normal tools like Excel or computers to handle. Therefore a special tech to store, process and to analyze it would be needed.

Example/~~scenerion~~ Scenario

City of Joburg - uses thousands of CCTV cameras + traffic ~~sensor~~ sensors + social media complaints to predict where potholes or protests will block traffic. ~~walk~~ And how to avoid that

6) Descriptive Analytics - it helps identify patterns, trends, and relationships within data. Clearly indicating "what happened" and "what is currently happening".

Example/~~Scenerion~~ Scenario

- Midrand taxi rank kota stall -
- Descriptive analytics → business sells mostly between 5-6pm. Average 22 kota's per hour
- What it tells you → when to schedule more staff or when all staff should be on the floor to enable more sales by ~~quet~~ quicker buying and leaving of customers. less long quees

7) Diagnostic Analytics - using data to investigate the causes of past outcomes. It works with descriptive data to find patterns, and reasons behind trends or irregularities.

Scenario - Spaza shop in Alfredgeville

- Revenue went down by R5,000 in April (Descriptive output)
- Supplier delivered bread ~~6 times~~ late 6 time in 4 weeks and there is a new shop opened on the backstreet (Diagnostic perception)

8) Predictive Analytics – "what could be the outcome or result." using statistical models, machine learning and historical data to estimate future outcomes, based on probability not certainties.

9) Prescriptive Analytics = ("what should be done")
prescriptive tells you what action to take to get best outcome. This is done by using data, predictions and business rules to suggest specific actions.

Example – Google Maps/Waze – they don't just predict ~~it predicts~~ traffic → also prescribes ("take M1, not M1, to save 12min")

10) Data Literacy – it is the ability to peruse, understand, use and communicate with data for better/~~informed~~ informed decision-making.

Example - Scenario – Spaza shop in Tembisa, hears that average SA household spends R200 on ~~bread~~ bread per month, and increases stock to 1000 loave baked

OR Bakery in Tembisa

Reality – his area/location has 30 houses, mostly old people/pensioners. He should baked 200 loaves because now he's stuck with 800 x rotten/stale